

Advanced Calculus I

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1 Preface

The notes that follow were used in the Fall 1999 and Spring 2000 semesters. The form in which they appear includes problems that “happened” during the course. Those problems or theorems that have a student’s name after them came consequent to student presentations. The labeled theorems were proven by students in the context of arguing another problem. The labeled problems are typically the result of articulating what was necessary to validate the argument the student was attempting to make.

I have left the notes in this form to provide an example showing how course notes evolve in response to what the students do in pursuit of the theorems you really want them to get. If one were to read the notes from another offering of this course sequence immediately prior or subsequent to this one, one would see a very different end-product.

This particular class lost its “star” student for the spring semester¹, yet was still populated by some good minds and diligent students. They got some nice results, but became stuck on Problem XVI. The class had proven all of the theorems that had led to the fall of Problem XVI in previous semesters, but this time it did not fall until the final examination. Most of the problems after Problem XVI are designed to lead the students to take fresh looks at the structures they had in place.

If a reader were to be interested, I would be glad to provide other “end-of-the-course” notes that started out with the same problem sequence as these, but followed quite different paths.

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¹He was studying abroad. When he returned in the fall, working alone for half a semester, he got the problems that the entire class had taken the entire spring semester to get.

2 Introduction

In this course, not only will you be responsible for understanding why the mathematics we cover is correct, but the responsibility for discovery will also be assigned to the class. One of the immediate results of this responsibility for doing mathematics yourself, rather than just learning how someone else did it, will likely be an acute awareness of the difference between the challenge associated with understanding why something is correct and discovering for yourself whether or not a conjecture is a theorem.

Doing mathematics can be extremely exhilarating when one succeeds in the discovery process; failing to do mathematics when one is putting in the time trying to do mathematics can be extremely frustrating. This introduction is intended to alert you to some tips which are designed to optimize the chances for success.

First, you must put in the time necessary to give your creative intelligence a chance to work. Flashes of insight typically occur after information is organized and mulled over. Commitment to solving problems often leads to help from the subconscious. Students often tell me that they got “the big idea” while walking across campus or after turning in for the night.

Second, solutions to problems need not come all at once. You may need to solve many small problems on the way to proving a theorem or disproving an incorrect conjecture. Some of the most important work in mathematics is the creation of technique. Take pride in progress toward a goal as well as reaching the goal. Any information you uncover is more than you knew before, and solving a problem is usually just a matter of putting together enough small solutions to allow you to see why the big problem is correct.

Students often tell me that they would be glad to put in the time if they just knew where to start. The following scheme is offered toward that end.

The awareness stage

1. Identify all the words in the problem, and make sure that you *know* the definition of each of them. Try to recall examples that have dealt with these notions before. If a definition is new, make some examples for the definition.
2. Identify any theorems that may have already dealt with ideas present in the problem. Put techniques that gave rise to proofs in those contexts firmly in mind.

The direct approach

3. Make an example which models the hypothesis to the problem, and try to show that the example exhibits the properties of the conclusion. (If you can prove that your example fails to have the properties of the conclusion, you will have shown that the problem is not a theorem!)

4. See if what allowed you to establish the conclusion in the example is a property of all examples covered by the hypothesis. If it is, write a proof. If not, ...
5. ... make an example which models the hypothesis, but fails to have whatever special properties you used to get the conclusion in the previous example. Go to 3.

The indirect, or contrapositive, approach

6. Suppose that the conclusion is false, and try to show that the hypothesis must be false as well. If the problem is not a theorem, any conclusions you get must be qualities which an example that disproves the conjecture must have.
7. Try to be aware of properties that, if they were added to the hypothesis, would guarantee the conclusion. Alternatively, you might also try to find conclusions that follow from the hypothesis, even if they do not include the one you seek. Even if you are not able to solve the problem as stated, you may be able to create a substitute theorem.

The main mindset is to be aware that even when arguments do not come quickly or easily, the hunt itself may be an important learning experience. Working on problems yourself is the central ingredient. Not only will it provide you with theorems that are “your own”, but even when someone beats you to a solution, it will put you in a much stronger position to analyze the argument given.

3 A Brief Review of the Logic of Quantification

1. There are two types of quantifications: universal and existential.
2. The most common syntax for an existential quantification is “There is ... so that ...”
3. The most common syntax for a universal quantification is “If ... then ...” (*This is also the syntax for conditionals!*)
4. Quantifications are statements, but the sentences they connect are not.
5. The sentences connected in a quantification describe properties.
6. The sentence following “There is” or “If” in a quantification, is called the sentence of inclusion. The sentence following “so that” or “then” is called the sentence of property.
7. An existential quantification asserts the existence of at least one object about which the properties of the sentences of inclusion and property are true.
8. A universal quantification asserts that any object about which the properties from the sentence of inclusion are true, must also have the properties from the sentence of property be true about it.
9. To prove directly that an existential quantification is true, you must exhibit an object about which the properties from the sentences of inclusion and property are true.
10. The negation of a universal quantification is the existential quantification with the same sentence of inclusion, and whose sentence of property is the negation of that of the universal.
11. The negation of an existential quantification is the universal quantification with the same sentence of inclusion, and whose sentence of property is the negation of the existential.

4 A Theory of Sets and Ordered Pairs

We will not create an axiomatic set theory. Following, however, is an idiomatic presentation of the conventions that axiomatic set theory implies, and presupposes the existence of formal English as a language for expressing properties.

The undefined terms are *set*, *element*, *ordered pair*, *first coordinate*, and *second coordinate*.

- i. A set consists of an element or elements.
- ii. An element of a set, and the set consisting of that element, are different objects.
- iii. A set is defined by stating the properties its elements have (or the property its element has).
- iv. Given a definition for a set, any object having the properties specified is an element of the set; and any element of the set has the properties specified in the definition.
- v. An ordered pair consists of a first coordinate and a second coordinate.
- vi. The first coordinate of an ordered pair may be the same set-theoretic object as the second coordinate, but as a part of the ordered pair, being the first coordinate is distinguishable from being the second coordinate.

We reserve a notation for the creation of definitions of sets and for defining ordered pairs.

Reserved symbols for definitions of sets are $\{ : \}$. A symbol is created to follow the open brace and precede the colon, and then properties that an element must have are stated in terms of that symbol, after the colon and before the closed brace. Thus

$$\{x : x \text{ is a number and } x > 5\}$$

stands for “the set to which an element belongs provided that it is a number and it is greater than 5”.

Reserved symbols for definitions of ordered pairs are $(,)$. The first coordinate of the ordered pair is written after the open parenthesis and before the comma; the second coordinate of the ordered pair is written after the comma and before the closed parenthesis. Thus $(p, 5)$ stands for the ordered pair whose first coordinate is p and whose second coordinate is 5.

5 Problems and Theorems Sequence

Definition 1: Suppose that each of A and B is a set. The statement that A is a *subset* of B means that if x is an element of A , then x is an element of B .

Notation: Suppose that each of A and B is a set. $A \subset B$ stands for “ A is a subset of B ”.

Definition 2: Suppose that each of A and B is a set, and that there is an element of A which is an element of B . The *intersection* of A with B is $\{p : p \text{ is an element of } A \text{ and } p \text{ is an element of } B\}$.

Notation: Suppose that each of A and B is a set. $A \cap B$ stands for “the intersection of A with B ”.

Definition 3: Suppose that each of A and B is a set. The *union* of A with B is $\{p : p \text{ is an element of } A \text{ or } p \text{ is an element of } B\}$.

Notation: Suppose that each of A and B is a set. $A \cup B$ stands for “the union of A with B ”.

Definition 4: Suppose that each of X and Y is a set, and

$$f = \{(p, q) : p \text{ is an element of } X \text{ and } q \text{ is an element of } Y\}$$

The statement that f is a *function* from X into Y means that

- (i.) if w is an element of X , then there is an element of f whose first coordinate is w ; and
- (ii.) if r and s are elements of f , then the first coordinate of r is not the same as the first coordinate of s .

Notation: If (p, q) is an element of the function f , then q may be written as $f(p)$.

Problem 1: Suppose that X is a set, and that Y is a set with more than one element. Show that

$$\{(p, q) : p \text{ is an element of } X \text{ and } q \text{ is an element of } Y\}$$

is not a function from X into Y .

Following are a set of axioms that describe the numbers.

Axiom 1: $\mathfrak{R}$ is a set.

$\mathfrak{R}$ will be referred to as “the numbers”, and an element of $\mathfrak{R}$ will be called a number.

We have a course sequence devoted to algebraic structure; hence we will offer proof for algebraic phenomena only on demand. Nevertheless, for completeness' sake, the axioms for the algebra of the numbers follow.

Axiom II (Algebra of the Numbers): “+” and “*” are functions from

$\{(x, y) : x \text{ is a number and } y \text{ is a number}\}$ into $\mathfrak{R}$ so that

- (i.) if each of x and y is a number, then $+(x, y) = +(y, x)$ and $*(x, y) = *(y, x)$;
- (ii.) if each of x, y , and z is a number, then

$$+((x, +(y, z))) = +((+(x, y), z))$$

and

$$*((x, *(y, z))) = *(((*(x, y), z))$$

- (iii.) 0 is a number so that if x is a number, then $+(0, x) = x$; and 1 is a number so that if x is a number, then $*(1, x) = x$;
- (iv.) if x is a number, then there is exactly one number, call it y , so that $+(x, y) = 0$; and if x is a number different than 0, then there is exactly one number, call it z , so that $*(x, z) = 1$; and
- (v.) if each of x, y , and z is a number, then

$$*((x, +(y, z))) = +((*(x, y), *(x, z)))$$

Notation: In the popular press, $+(x, y)$ is usually written as $x + y$, and $*(x, y)$ is usually written as $x * y$. Feel free to use this notation.

Definition 5 (Geometry of the Numbers): Suppose that X is a set, and that $P \subset \{(x, y) : x \text{ is an element of } X, \text{ and } y \text{ is an element of } X\}$. The statement that P is an *order* on X means that

- (i.) if x is an element of X , then (x, x) is not an element of P ;
- (ii.) if x and y are elements of X , then (x, y) is an element of P or (y, x) is an element of P ; and
- (iii.) if (x, y) and (y, z) are elements of P , then (x, z) is an element of P .

Notation: If P is an order on X , xPy means that (x, y) is an element of P .

Axiom III: “<” is an order on $\mathfrak{R}$.

Problem 2: Suppose that X is a set, and P is an order on X . Show that P is not a function from X into X .

Definition 6: Suppose that $X \subset \mathfrak{R}$, and p is a number. The statement that p is the *maximum* of X means that p is an element of X , and if q is an element of X different than p , then $q < p$. The statement that p is the *minimum* of X means that p is an element of X , and if q is an element of X different than p , then $p < q$.

Axiom IV: If M and Q are subsets of $\mathbb{R}$ so that

- (i.) $M \cup Q = \mathbb{R}$; and
- (ii.) if x is an element of M , and y is an element of Q , then $x < y$;

then either M has a maximum and Q has no minimum, or M has no maximum and Q has a minimum.

Definition 7: Suppose that X is a set. The statement that f is a *sequence* in X means that f is a function from the natural numbers into X .

Axiom V: There is a sequence in $\mathbb{R}$, call it f , so that if x and y are numbers so that $x < y$, then there is a natural number, call it n , so that $x < f(n)$ and $f(n) < y$.

Axiom VI (The Combining Axiom): Suppose that x and y are numbers so that $x < y$, and z is a number. Then $+(x, z) < +(y, z)$.

Definition 8: Suppose that a and b are numbers, and $a < b$. The *segment* from a to b is the set to which a number belongs, provided that a is less than it, and it is less than b . The *interval* from a to b is the set to which a number belongs, provided that it is an element of the segment from a to b , or it is a , or it is b .

Notation: (a, b) is read “the segment from a to b ”, and $[a, b]$ is read “the interval from a to b ”.

Definition 9: Suppose that $X \subset \mathbb{R}$, that f is a function from X into $\mathbb{R}$, and p is an element of X . The statement that f is *continuous at p* means that if V is a segment so that $f(p)$ is an element of V , then there is a segment, call it U , so that p is an element of U , and if q is an element of $U \cap X$, then $f(q)$ is an element of V . The statement that f is *continuous on X* means that if p is an element of X , then f is continuous at p .

Problem 3: Suppose that each of A and B is a number, and that

$$L = \{(x, y) : x \text{ is a number and } y = (A * x) + B\}$$

Show that L is a function from $\mathbb{R}$ into $\mathbb{R}$ so that L is continuous on $\mathbb{R}$.

Problem 4: Suppose that $Q = \{(x, y) : x \text{ is a number and } y = 2 * x\}$. Show that Q is a function from $\mathbb{R}$ into $\mathbb{R}$ so that Q is continuous on $\mathbb{R}$.

Problem 5: Suppose that X is a subset of $\mathbb{R}$. Show that there is a function from X into $\mathbb{R}$ that is continuous on X .

Definition 10: Suppose that $X \subset \mathbb{R}$, and that each of f and g is a function from X into $\mathbb{R}$. The *sum* of f with g is

$$\{(p, q) : p \text{ is an element of } X \text{ and } q = f(p) + g(p)\}$$

Notation: $f+g$ may be read “the sum of f with g ”.

Definition 11: Suppose that $X \subset \mathbb{R}$, and that each of f and g is a function from X into $\mathbb{R}$. The *product* of f with g is

$$\{(p, q) : p \text{ is an element of } X \text{ and } q = f(p) * g(p)\}$$

Notation: $f*g$ may be read “the product of f with g ”.

Problem 6: Suppose that $X \subset \mathbb{R}$, that p is an element of X , and that each of f and g is a function from X into $\mathbb{R}$ so that f is continuous at p and g is continuous at p . Show that $f+g$ is continuous at p .

Problem 7: Suppose that $X \subset \mathbb{R}$, that each of f and g is a function from X into $\mathbb{R}$, and that p and q are elements of X . Show that $f*g(p) + -f*g(q) = f(p) * (g(p) + -g(q)) + g(q) * (f(p) + -f(q))$.

Problem 8: Suppose that $X \subset \mathbb{R}$, that p is an element of X , and that each of f and g is a function from X into $\mathbb{R}$ so that f is continuous at p and g is continuous at p . Show that $f*g$ is continuous at p .

Definition 12: Suppose that each of X , Y , and Z is a set, that f is a function from X into Y , and that g is a function from Y into Z . The *composition* of g with f is $\{(p, q) : p \text{ is an element of } X \text{ and } q = g(f(p))\}$.

Notation: $g \circ f$ may be read “the composition of g with f ”.

Problem 9: Suppose that each of X , Y , and Z is a subset of $\mathbb{R}$, that f is a function from X into Y that is continuous on X , and that g is a function from Y into Z that is continuous on Y . Show that $g \circ f$ is continuous on X .

Problem 10: Suppose that X is a subset of $\mathbb{R}$, that f is a function from X into $\mathbb{R}$ so that f is continuous on X , that p is an element of X , and

$$SAfp = \{(x, y) : x \text{ is an element of } X, \text{ and } x \text{ is not } p, \\ \text{and } y = \frac{f(x) + -f(p)}{x + -p}\}$$

Show that $SAfp$ is continuous on

$$\{x : x \text{ is an element of } X, \text{ and } x \text{ is not } p\}$$

Theorem 1.5 (Liacouras): Suppose that X is a set, and P is an order on X . Then X has more than one element.

Theorem 1.95 (Penny): Suppose that X is a set, and P is an order on X so that there are elements of P so that the first coordinate of one is the second coordinate of the other. Then P is not a function from X into X .

Technical Lemma 1 (in the proof of Theorem 7 (Enge)): Suppose that each of x and y is a number. Then $-(x * y) = (-x) * y$.

Technical Lemma 2 (in the proof of Theorem 3 (Robinson)):

Suppose that x and y are numbers so that $x < y$, and A is a number so that $A > 0$. Then $A * x < A * y$.

Technical Lemma 2' (in the proof of Theorem 3 (Robinson)):

Suppose that x and y are numbers so that $x < y$, and A is a number so that $A < 0$. Then $A * y < A * x$.

Theorem 4.9 (from discussion triggered by a question of Penney):

Suppose that f is a sequence in $\mathfrak{R}$. Then f is continuous on N .

Problem 11: Suppose that

$$\Gamma = \left\{ (x, y) : x \text{ is a number different than } 0, \text{ and } y = \frac{1}{x} \right\}$$

Show that Γ is continuous on

$$\{x : x \text{ is a number different than } 0\}$$

Problem 12: Suppose that

$$\Gamma = \left\{ (x, y) : x \text{ is a number different than } 0, \text{ and } y = \frac{1}{x} \right\}$$

and that p is a number. Show that $\Gamma \cup \{(0, p)\}$ is not continuous at 0.

Technical Lemma 3 (Bellino): Suppose that A and B are numbers, and $0 < A$ and $A < B$. Then $A^2 < B^2$.

Technical Lemma 3' (Bellino): Suppose that A and B are numbers, and $B < A$ and $A < 0$. Then $A^2 < B^2$.

Technical Lemma 4 (Liacouras): Suppose that A and B are numbers so that $0 < A$ and $A < B$. Then $\frac{1}{A} > \frac{1}{B}$.

Technical Lemma 4' (Liacouras): Suppose that A and B are numbers so that $A < 0$ and $B < A$. Then $\frac{1}{A} > \frac{1}{B}$.

Definition 13: Suppose that $M \subset \mathfrak{R}$. The statement that M is *bounded* means that there is a segment, call it S , so that $M \subset S$.

Theorem 12.1 (from Dickson's proof of Theorem 12): Suppose that $a < 0$ and $b > 0$. Then the range of

$$\left\{ (x, y) : x \text{ is an element of } \underline{(a, b)} \text{ and } x \neq 0, \text{ and } y = \frac{1}{x} \right\}$$

is not bounded.

Problem 13: Suppose that a and b are numbers, and that f is a function from $[a, b]$ into $\mathbb{R}$ so that f is continuous on $[a, b]$. Show that $\mathcal{R}_f$ is bounded.

Definition 14: Suppose that $M \subset \mathbb{R}$, and that f is a function from M into $\mathbb{R}$. The statement that f has the *property of Lipschitz* means that there is a number, call it k , so that if x and y are elements of M , then

$$|f(x) - f(y)| \leq k * |x - y|$$

Problem 14: Suppose that each of A and B is a number, and

$$L = \{(x, y) : x \text{ is a number and } y = (A * x) + B\}$$

Show that L has the property of Lipschitz.

Problem 15: Suppose that

$$Q = \{(x, y) : x \text{ is a number and } y = x^2\}$$

Show that Q has the property of Lipschitz.

Problem 16: Suppose that $M \subset \mathbb{R}$, and that f is a function from M into $\mathbb{R}$ so that f has the property of Lipschitz. Show that f is continuous on M .

Technical Lemma 5 (Enge): Suppose that A, B, C , and D are numbers so that $A < B$ and $C < D$. Then $A + C < B + D$.

Theorem 5.9 (Enge): Suppose that A, B, x , and y are numbers so that

$A < x + y$ and $x + y < B$. Then there are numbers, call them p, q, r , and s , so that $p + q = A, r + s = B, x$ is an element of $\underline{(p, r)}$, and y is an element of $\underline{(q, s)}$.

Definition 15: Suppose that $X \subset \mathbb{R}$, and f is a function from X into $\mathbb{R}$. The *range* of f is

$$\{x : \text{there is an element of } f \text{ so that } x \text{ is the second coordinate of it}\}$$

Notation: $\mathcal{R}_f$ stands for “the range of f ”.

Problem 17: Suppose that $X \subset \mathbb{R}$, that f is a function from X into $\mathbb{R}$ so that f is continuous on X , and that $Y \subset X$. Show that

$$\{(x, y) : x \text{ is an element of } Y \text{ and } y = f(x)\}$$

is continuous on Y .

Definition 16: Suppose that $X \subset \mathbb{R}$, that f is a function from X into $\mathbb{R}$, that w is an element of X , that p is a number, and

$$SAfw = \left\{ (x, y) : x \in X \text{ and } y = \frac{f(x) - f(w)}{x - w} \right\}$$

The statement that p is the *derivative* of f at w means that $SAfw \cup \{(w, p)\}$ is continuous at w . The derivative of f is

$$\{(x, y) : x \text{ is an element of } X, \text{ and } y \text{ is the derivative of } f \text{ at } x\}$$

Problem 18: Show that $\{(x, y) : x \text{ is a number and } y = 2 * x\}$ is the derivative of $\{(x, y) : x \text{ is a number and } y = x^2\}$.

Problem 19: Suppose that $X \subset \mathbb{R}$, that f is a function from X into $\mathbb{R}$, that w is an element of X , and that p is the derivative of f at w . Show that f is continuous at w .

Technical Lemma 6 (Dostert): Suppose that each of x and y is a number. Then $|x * y| = |x| * |y|$.

Theorem 15' (Bellino): Suppose that X is a bounded subset of $\mathbb{R}$, and that $Q' = \{(x, y) : x \text{ is an element of } X \text{ and } y = x^2\}$. Then Q' has the property of Lipschitz.

Problem 20: Suppose that $M \subset \mathbb{R}$, and that M is bounded. Show that

$$\{x : \text{if } p \text{ is an element of } M, \text{ then } p \leq x\}$$

has a minimum.

Problem 21: Suppose that $X \subset \mathbb{R}$, and f is a function from X into $\mathbb{R}$ so that f has the property of Lipschitz. Show that

$$\{p : \text{if } x \text{ and } y \text{ are elements of } X, \text{ then } |f(x) - f(y)| \leq p * |x - y|\}$$

has a minimum.

Problem 22: Suppose that $X \subset \mathbb{R}$, and that f is a function from X into $\mathbb{R}$ so that f has the property of Lipschitz. Show that R_f is bounded.

Problem 23 (Liacouras): Suppose that f is a function from $[a, b]$ into $\mathbb{R}$. Show that R_f has a maximum and R_f has a minimum.

Theorem 12.1 (from Dickson's proof of Theorem 12): Suppose that $a \leq 0$ and $b > 0$. Then the range of

$$\left\{ (x, y) : x \text{ is an element of } \underline{(a, b)} \text{ and } x \neq 0, \text{ and } y = \frac{1}{x} \right\}$$

is not bounded.

Problem 24: Suppose that X is a subset of $\mathbb{R}$, that f is a function from X into $\mathbb{R}$, and that each of x and p is a number so that x is not an element of X and $f \cup \{(x, p)\}$ is continuous at x . Show that if q is a number different than p , then $f \cup \{(x, q)\}$ is not continuous at x .

Problem 22': Suppose that $X \subset \mathbb{R}$, that X is bounded, and that f is a function from X into $\mathbb{R}$ so that f has the property of Lipschitz. Show that R_f is bounded.

Problem 23': Suppose that f is a function from $[a, b]$ into $\mathbb{R}$ so that f is continuous on $[a, b]$. Show that R_f has a maximum and R_f has a minimum.

Problem 25: Suppose that s is a sequence in

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = \underline{a, b}\}$$

so that if n is a natural number, then $s(n+1) \subset s(n)$. Show that there is a number, call it x , so that if n is a natural number, then x is an element of $s(n)$.

Problem 26: Suppose that s is a sequence in

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = [a, b]\}$$

so that if n is a natural number, then $s(n+1) \subset s(n)$. Show that there is a number, call it x , so that if n is a natural number, then x is an element of $s(n)$.

Problem 27 (Penney): Suppose that f is a function from $[a, b]$ into $\mathbb{R}$ so that f is continuous on $[a, b]$, that $f(a) < f(b)$, and that c is an element of $\underline{(f(a), f(b))}$. Show that c is an element of R_f .

Theorem 22.5 (Penney): Suppose that $X \subset \mathbb{R}$; that f is a function from X into $\mathbb{R}$ so that R_f is bounded and has the property that if x and y are elements of R_f and z is an element of $\underline{(x, y)}$, then z is an element of R_f ; that

$$Y = \{p : \text{if } x \text{ is an element of } R_f, \text{ then } p < x\}$$

and that

$$Z = \{p : \text{if } x \text{ is an element of } R_f, \text{ then } p > x\}$$

Then Y has a max or R_f has a min, and R_f has a max or Z has a min.

Technical Lemma 7 (Enge): Suppose that w, x, y , and z are numbers so that $0 < w, 0 < x, 0 < y, 0 < z, w < x$, and $y < z$. Then $w * y < x * z$.

Theorem 7.8 (suggested by work of Dickson and proven by Enge):

Suppose that a, x, y , and b are numbers so that $a < x * y$ and $x * y < b$. Then there are numbers, call them A and B , so that $A * B = a$, $A < x$, and $B < y$; and there are numbers, call them C and D , so that $C * D = b$, $x < C$, and $y < D$.

Theorem 7.9 (Enge): Suppose that $X \subset \mathbb{R}$, f is a function from X into $\mathbb{R}$, p is a number so that f is continuous at p , and c is a number. Then $\{(x, y) : x \text{ is an element of } X \text{ and } y = c * f(x)\}$ is continuous at p .

Problem 28: Suppose that t is a sequence in $[A, B]$ so that if n and k are natural numbers, then $t(n)$ is not $t(k)$. Show that there is an element of $[A, B]$, call it p , so that if S is a segment so that p is an element of S , then there is an element of R_t which is an element of S and different than p .

Problem 29: Suppose that f is a function from $[A, B]$ into $\mathfrak{R}$, so that f' is a function from $[A, B]$ into $\mathfrak{R}$ which is continuous on $[A, B]$. Show that f has the property of Lipschitz.

Problem 30: Suppose that s is a sequence in

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = \underline{a, b}\}$$

so that

- (i.) if n is a natural number and $s(n) = \underline{a, b}$, then a is not an element of $s(n+1)$, b is an element of $s(n+1)$, and b is not an element of $s(n+2)$; and
- (ii.) there is a number, call it x , so that if p is an element of $\bigcup_{n \in \mathbb{N}} s(n)$, then $p < x$.

Show that there is a number, call it U , so that if S is a segment of which U is an element, then there is a natural number, call it K , so that if $n > K$, then $s(n)$ is a subset of S .

Problem 31: Suppose that t is a function from $[A, B]$ into

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = \underline{a, b}\}$$

so that if x is an element of $[A, B]$, then x is an element of $t(x)$. Show that there is a natural number, call it K , and a function from

$$\{x : x \text{ is a natural number and } x \leq K\}$$

into the range of t , call it t' , so that $[A, B] \subset \bigcup_{n \leq K} t'(n)$.

Theorem 19.8 (Penny): Suppose that M is a subset of $\mathfrak{R}$ so that M is bounded, that

$$X = \{p : \text{if } x \text{ is an element of } M, \text{ then } x \leq p\}$$

and that there is an element of M which is an element of X . Then there is only one element of M which is an element of X , and it is the minimum of X .

Problem 32 (Dickson): Suppose that a and b are numbers so that $a < b$, and that f is a function from $[a, b]$ into $\mathfrak{R}$ so that f is continuous on $[a, b]$ and $f(a) < f(b)$. Show that there are numbers, call them p and q , so that $[p, q]$ is a subset of $[a, b]$, and if c and d are elements of $[p, q]$ so that $c < d$, then $f(c) < f(d)$.

Problem 33: Suppose that f is a function from $\mathfrak{R}$ into $\mathfrak{R}$ so that the range of f is bounded, and that

$$g = \{(x, y) : x \text{ is a number and } y = x * f(x)\}$$

Show that g is continuous at 0.

Problem 34: Suppose that f is a function from $[a, b]$ into $\mathfrak{R}$ so that f is continuous on $[a, b]$, that $f(a) < f(b)$, that c is a number in $\underline{(f(a), f(b))}$, that

$$P = \{x : x \leq a; \text{ or } x > a \text{ and if } p \text{ is an element of } [a, x], \text{ then } f(p) < c\}$$

and that

$$Q = \{x : x \text{ is a number and } x \text{ is not an element of } P\}$$

Show that

- (i.) $P \cup Q = \mathfrak{R}$; and
- (ii.) if x is an element of P and y is an element of Q , then $x < y$.

Problem 35: Suppose that s is a sequence in $[a, b]$, and c and d are numbers so that c is an element of $\underline{(a, b)}$ and d is an element of $\underline{(c, b)}$. Show that $s(1)$ is not an element of one of $\underline{[a, c]}$, $\underline{[c, d]}$, and $\underline{[d, b]}$.

Problem 36: Suppose that s is a sequence in $[a, b]$. Show that R_s is not $[a, b]$.

Problem 25': (Robinson) Suppose that s is a sequence in

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = \underline{(a, b)}\}$$

so that

- (i.) if n is a natural number, then $s(n+1) \subset s(n)$; and
- (ii.) if $s(n) = \underline{(c, d)}$ and $s(n+1) = \underline{(e, m)}$, then $e > c$ and $d < m$.

Show that there is a number, call it x , so that if n is a natural number, then x is an element of $s(n)$.

Problem 37: Suppose that f is a function from $[A, B]$ into $\mathfrak{R}$ so that f is continuous on $[A, B]$. Show that f has the property of Lipschitz.

Definition 16: Suppose that A and B are numbers so that $A < B$, and n is a natural number. The statement that P is a *partition* of $[a, b]$ of size n means that P is a function from $\{k : k \text{ is a natural number and } k \leq n\}$ into $[A, B]$ so that

- (i.) $P(1) = A$ and $P(n) = B$; and
- (ii.) if k is a natural number less than n , then $P(k) < P(k+1)$.

Definition 17: Suppose that P is a partition of $[A, B]$ of size n . The statement that Q is an *interpolating sequence* for P means that Q is a function from

$$\{k : k \text{ is a natural number and } k < n\}$$

into $[A, B]$ so that if k is a natural number less than n , then $Q(k)$ is an element of $[P(k), P(k+1)]$.

Definition 18: Suppose that P is a partition of $[A, B]$ of size n , that Q is an interpolating sequence for P , and that f is a function from $[A, B]$ into $\mathfrak{R}$. The

$$\text{Riemann sum for } f \text{ by } P \text{ and } Q \text{ is } \sum_{k=1}^{n+1} f(Q(k)) * (P(k+1) - P(k)).$$

Notation: $R_{f,P,Q}$ stands for “the Riemann sum for f by P and Q ”.

Problem 38: Suppose that f is a function from $[A, B]$ into $\mathfrak{R}$, so that f is continuous on $[A, B]$ and that P is a partition of $[A, B]$ of size n . Show that

$$\{m : \text{there is an interpolating sequence for } P, \text{ call it } Q, \\ \text{so that } m = R_{f,P,Q}\}$$

has a maximum and a minimum.

6 Final Examination

You may use your class notes for this examination, but should not consult any human sources other than your professor. In writing your solutions, remember that you will not be present to explain what you meant to say; be precise and be correct. Definitions and notations are those in the course notes. In making arguments, you may cite problems proven during the course, provided that their proofs do not depend on the problem being argued.

Part I

Do one problem from each of the following five sets.

1.

- (a.) Suppose that each of P and Q is a number, and that

$$f = \{(x, y) : x \text{ is a number and } y = (P * x) + Q\}$$

Show that f is continuous on $\mathbb{R}$.

- (b.) Show that

$$\{(x, y) : x \text{ is a number and } y = x^2\}$$

is continuous on $\mathbb{R}$.

- (c.) Show that

$$\left\{ (x, y) : x \text{ is different than } 0 \text{ and } y = \frac{1}{x} \right\}$$

is continuous on $\mathbb{R}$.

2.

- (a.) Suppose that X is a subset of $\mathbb{R}$, that p is an element of X , and that each of f and g is a function from X into $\mathbb{R}$ so that each of f and g is continuous at p . Show that $f + g$ is continuous at p .

- (b.) Suppose that X is a subset of $\mathbb{R}$, that p is an element of X , and that each of f and g is a function from X into $\mathbb{R}$ so that each of f and g is continuous at p . Show that $f * g$ is continuous at p .

- (c.) Suppose that each of X and Y is a subset of $\mathbb{R}$, that f is a function from X into Y and g is a function from Y into $\mathbb{R}$, and that p is an element of X so that f is continuous at p , and g is continuous at $f(p)$. Show that $g \circ f$ is continuous at p .

3.

(a.) Suppose that q is a number. Show that

$$\left\{ (x, y) : x \text{ is a number different than } 0 \text{ and } y = \frac{1}{x} \right\}$$

is not continuous at 0.

(b.) Show that

$$\{(x, y) : x \text{ is a number and } y = 2 * x\}$$

is the derivative of

$$\{(x, y) : x \text{ is a number and } y = x^2\}$$

4.

(a.) Suppose that X is a subset of $\mathfrak{R}$ so that X is bounded. Show that

$$\{p : \text{if } q \text{ is an element of } X, \text{ then } p \leq q\}$$

has a maximum.

(b.) Suppose that s is a sequence in

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = [a, b]\}$$

so that if n is a natural number, then $s(n+1)$ is a subset of $s(n)$.
 Show that there is a number, call it p , so that if n is a natural number,
 then p is an element of $s(n)$.

(c.) Suppose that X is a subset of $\mathfrak{R}$, and that f is a function from X into $\mathfrak{R}$ so that f has the property of Lipschitz. Show that f is continuous on X .(d.) Suppose that X is a subset of $\mathfrak{R}$, that f is a function from X into $\mathfrak{R}$, and p is an element of X , and q is a number so that q is the derivative of f at p . Show that f is continuous at p .

5.

(a.) Suppose that a and b are numbers so that $a < b$, and f is a function from $[a, b]$ into $\mathfrak{R}$ so that f is continuous on $[a, b]$. Show that the range of f is bounded.(b.) Suppose that a and b are numbers so that $a < b$, that f is a function from $[a, b]$ into $\mathfrak{R}$ so that f is continuous on $[a, b]$, that $f(b) < f(a)$, and that c is an element of $(f(b), f(a))$. Show that c is an element of the range of f .(c.) Suppose that a and b are numbers so that $a < b$, and f is a function from $[a, b]$ into $\mathfrak{R}$ so that f is continuous on $[a, b]$. Show that the range of f has a minimum.

Part II

1.

- (a.) Suppose that s is a sequence in $[a, b]$ so that if n is not m , then $s(n)$ is not $s(m)$; and

$$X = \{p : p < a; \text{ or there is a natural number, call it } n, \\ \text{so that if } m > n, \text{ then } p < s(m)\}$$

and

$$Y = \{p : p \text{ is not an element of } X\}$$

Show that $X \cup Y = \mathfrak{R}$, and if A is an element of X and B is an element of Y , then $A < B$.

- (b.) Suppose that s is a sequence in $[a, b]$ so that if n is not m , then $s(n)$ is not $s(m)$. Show that there is a number in $[a, b]$, call such a number q , so that if S is a segment so that q is an element of U , then there is an element of the range of s different than q which is an element of U .

2.

- (a.) Suppose that w, x, y , and z are numbers so that $w < x, x < y$, and $y < z$. Show that there is an interval, call it M , so that M is a subset of $\underline{(w, z)}$, and $\underline{(x, y)}$ is a subset of M .
- (b.) Suppose that s is a sequence in

$$\{m : \text{there are numbers, call them } a \text{ and } b, \text{ so that } m = \underline{(a, b)}\}$$

so that if n is a natural number and $s(n) = \underline{(a, b)}$ and $s(n+1) = \underline{(c, d)}$, then $a < c$ and $d < b$. Show that there is a number, call it p , so that if n is a natural number, then p is an element of $s(n)$.

3.

- (a.) Suppose that s is a sequence in $[a, b]$, and n is a natural number. Show that there is an interval, call it M , so that M is a subset of $[a, b]$, and if k is a natural number less than n , then $s(k)$ is not an element of M .
- (b.) Suppose that s is a sequence in $[a, b]$. Show that the range of s is not $[a, b]$.

4. Do Problem 38:

Suppose that f is a function from $[A, B]$ into $\mathfrak{R}$, so that f is continuous on $[A, B]$ and that P is a partition of $[A, B]$ of size n . Show that

$$\{m : \text{there is an interpolating sequence for } P, \text{ call it } Q, \\ \text{so that } m = R_{f,P,Q}\}$$

has a maximum and a minimum.

Part III

1.

$D = \left\{ x : x \text{ is an element of } [0, 1], \text{ and there is an odd integer, call it } k, \right.$
 $\left. \text{and a natural number, call it } n, \text{ so that } x = \frac{k}{2^n} \right\}$

$RL = \{(x, y) : x \text{ is an element of } [0, 1]; \text{ and if } x \text{ is an element of } D,$
 $\text{and } k \text{ is an odd natural number, and } n \text{ is a natural}$
 $\text{number, so that } x = \frac{k}{2^n}, \text{ then } y = \frac{1}{2^n};$
 $\text{or if } x \text{ is not an element of } D, \text{ then } y = 0\}$

- (a.) Show that there is an element of $[0, 1]$, call it p , so that RL is continuous at p .
- (b.) Show that there is an element of $[0, 1]$, call it p , so that RL is not continuous at p .
- (c.) Double credit for specifying the elements of

$$\{x : RL \text{ is continuous at } x\}$$

and proving that every element of your set has the property, or for showing that RL is not continuous at any element of $[0, 1]$.

2.

- (a.) Suppose that X is a subset of $\mathfrak{R}$ with more than one element, that p is an element of X , and each of f and g is a function from X into $\mathfrak{R}$. Show that $SAfp + SAgp = SA(f + g)p$.
- (b.) Suppose that X is a subset of $\mathfrak{R}$ with more than one element, that p is an element of X , and each of f and g is a function from X into $\mathfrak{R}$. Show that $SAfp * SAgp = g * SAfp + f(p) * SAgp$.
- (c.) Suppose that X is a subset of $\mathfrak{R}$, that each of f and g is a function from X into $\mathfrak{R}$, that $f'(p) = q$, and $g'(p) = s$. Show that $q + s$ is the derivative of $f + g$ at p , and that $q * g(p) + s * f(p)$ is the derivative of $f * g$ at p .

3.

- (a.) Suppose that X and Y are subsets of $\mathfrak{R}$ and both have more than one element, that f is a function from X into Y , that g is a function from Y into $\mathfrak{R}$, and that p is an element of X . Show that if x is an element of X different than p so that $f(p)$ is not $f(x)$, then $SA(g \circ f)p(x) = SAgf(p)(f(x)) * SAgp(x)$.

- (b.) Suppose that X and Y are subsets of $\mathfrak{R}$, that g is a function from Y into $\mathfrak{R}$, and that p is an element of X , and each of q and r is a number so that q is the derivative of f at p and r is the derivative of g at $f(p)$. Show that $r * q$ is the derivative of $g \circ f$ at p .